

## Machine Learning Lab

Enrolment No: \_\_\_\_\_

Division: \_\_\_\_\_

Roll No: \_\_\_\_\_

Name: \_\_\_\_\_

### Experiment No.: 09

**Title:** Write a program to demonstrate the working of the K means algorithm. Use an appropriate data set and visualize results

#### Theory:

##### Unsupervised learning :

Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labelled responses. The most common unsupervised learning method is cluster analysis, which is used for exploratory data analysis to find hidden patterns or grouping in data.

Common clustering algorithms include:

- Hierarchical clustering: builds a multilevel hierarchy of clusters by creating a cluster tree
- k-Means clustering: partitions data into k distinct clusters based on distance to the centroid of a cluster
- Gaussian mixture models: models clusters as a mixture of multivariate normal density components
- Self-organizing maps: uses neural networks that learn the topology and distribution of the data
- Hidden Markov models: uses observed data to recover the sequence of states

**Unsupervised learning methods** are used in bioinformatics for sequence analysis and genetic clustering; in data mining for sequence and pattern mining; in medical imaging for image segmentation; and in computer vision for object recognition.

##### Clustering and its types:

Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group than those in other groups. In simple words, the aim is to segregate groups

with similar traits and assign them into clusters.



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Let's understand this with an example. Suppose, you are the head of a rental store and wish to understand preferences of your costumers to scale up your business. Is it possible for you to look at details of each costumer and devise a unique business strategy for each one of them? Definitely not. But, what you can do is to cluster all of your costumers into say 10 groups based on their purchasing habits and use a separate strategy for costumers in each of these 10 groups. And this is what we call clustering.

A "clustering" is essentially a set of such clusters, usually containing all objects in the data set. Additionally, it may specify the relationship of the clusters to each other, for example, a hierarchy of clusters embedded in each other. Clustering's can be roughly distinguished as:

Hard clustering: each object belongs to a cluster or not

- Soft clustering (also: fuzzy clustering): each object belongs to each cluster to a certain degree (for example, a likelihood of belonging to the cluster)

### Types:

Strict partitioning clustering: each object belongs to exactly one cluster

- Strict partitioning clustering with outliers: objects can also belong to no cluster, and are considered outliers
- Overlapping clustering (also: alternative clustering, multi-view clustering): objects may belong to more than one cluster; usually involving hard clusters
- Hierarchical clustering: objects that belong to a child cluster also belong to the parent cluster
- Subspace clustering: while an overlapping clustering, within a uniquely defined subspace, clusters are not expected to overlap.

### K means algorithm

Kmeans algorithm is an iterative algorithm that tries to partition the dataset into K pre-defined distinct non-overlapping subgroups (clusters) where each data point belongs to only one group. It tries to make the inter-cluster data points as similar as possible while also keeping the clusters as different (far) as possible. It assigns data points to a cluster such that the sum of the squared distance between the data points and the cluster's centroid (arithmetic mean of all the data points that belong to that cluster)

is at the minimum. The less variation we have within clusters, the more homogeneous (similar) the data points are within the same cluster.

The way k means algorithm works is as follows:

1. Specify number of clusters K.
  2. Initialize centroids by first shuffling the dataset and then randomly selecting K data points for the centroids without replacement.
  3. Keep iterating until there is no change to the centroids. i.e assignment of data points to clusters isn't changing.
- Compute the sum of the squared distance between data points and all centroids.
  - Assign each data point to the closest cluster (centroid).
  - Compute the centroids for the clusters by taking the average of the all data points that belong to each cluster.

The approach k-means follows to solve the problem is called Expectation- Maximization.

#### Applications:

- Market segmentation
- Document clustering
- Image compression
- Customer behavior analysis

#### Advantages of K-Means Algorithm

1. **Simplicity** – Easy to understand and implement.
2. **Efficiency** – Works fast on large datasets compared to other clustering algorithms.
3. **Scalability** – Can handle datasets with many variables and large sample sizes.
4. **Versatility** – Works well for data that forms spherical clusters.
5. **Unsupervised Learning** – Does not require labeled data.
6. **Deterministic Nature (with k-means++)** – Improved centroid initialization helps avoid poor clustering.

#### Disadvantages of K-Means Algorithm

1. **Need to Predefine K** – The number of clusters must be known in advance, which is not always possible.
2. **Sensitive to Initialization** – Different initial centroid choices can give different results.
3. **Assumes Spherical Clusters** – Struggles with clusters of irregular

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>shapes, densities, or sizes.</p> <ol style="list-style-type: none"> <li><b>Outlier Sensitivity</b> – Outliers can distort centroids and clustering results.</li> <li><b>May Converge to Local Minima</b> – Does not guarantee finding the global optimum solution.</li> <li><b>Not Suitable for Categorical Data</b> – Works best with continuous numeric features.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pseudo code or Program</b> | <pre> import numpy as np import pandas as pd import urllib.request from sklearn.cluster import KMeans from sklearn.preprocessing import StandardScaler import matplotlib.pyplot as plt  # URL for the Wine dataset (CSV format) url = "https://archive.ics.uci.edu/ml/machine-learning-databases/wine/wine.data"  # Load the Wine dataset into a pandas DataFrame column_names = ['Class', 'Alcohol', 'Malic_Acid', 'Ash', 'Alcalinity_of_Ash',                 'Magnesium', 'Total_Phenols', 'Flavanoids', 'Nonflavanoid_phenols',                 'Proanthocyanins', 'Color_Intensity', 'Hue',                 'OD280/OD315_of_diluted_wine',                 'Proline'] wine_data = pd.read_csv(url, header=None, names=column_names)  # Step 1: Preprocess the data (Standardization) features = wine_data.drop('Class', axis=1) # Remove the target column scaler = StandardScaler() features_scaled = scaler.fit_transform(features)  # Step 2: Apply K-Means clustering (Let's assume we want 3 clusters) kmeans = KMeans(n_clusters=3, random_state=42) wine_data['Cluster'] = kmeans.fit_predict(features_scaled)  # Step 3: Visualize the clusters (using the first two features for simplicity) plt.figure(figsize=(8, 6)) plt.scatter(wine_data['Alcohol'], wine_data['Malic_Acid'],             c=wine_data['Cluster'], cmap='viridis') plt.xlabel('Alcohol') plt.ylabel('Malic Acid') plt.title('K-Means Clustering on Wine Dataset') </pre> |

|                |                                                                                                                                          |
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|                | <pre>plt.colorbar(label='Cluster') plt.show()  # Step 4: Evaluate Clusters print(wine_data.head())</pre>                                 |
| <b>Outputs</b> | <p>(Students can attach extra pages if necessary)</p>  |

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| <div style="border: 1px solid green; padding: 5px; text-align: center;"> <b>Machine Learning Lab</b><br/>         (Students can attach extra pages if necessary)       </div> |                                                                                                                                                                                                                                                                      |
| <b>Exercise for Practice</b>                                                                                                                                                  |                                                                                                                                                                                                                                                                      |
| <b>Q1.</b>                                                                                                                                                                    | What are the advantages and disadvantages of K-means Algorithm?                                                                                                                                                                                                      |
| <b>Ans:</b>                                                                                                                                                                   | <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="background-color: #800080; height: 20px; margin-bottom: 5px;"></div>       |
| <b>Q2.</b>                                                                                                                                                                    | What are some stopping criteria of K-means clustering?                                                                                                                                                                                                               |
| <b>Ans</b>                                                                                                                                                                    | <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> |
| <b>Q3</b>                                                                                                                                                                     | What is the difference between K-means and K-nearest neighbours?                                                                                                                                                                                                     |
| <b>Ans</b>                                                                                                                                                                    | <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> |
| <b>Q4.</b>                                                                                                                                                                    | Explain some cases where K-means clustering fails to give good results.                                                                                                                                                                                              |
| <b>Ans</b>                                                                                                                                                                    | <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px dashed black; height: 20px; margin-bottom: 5px;"></div> |



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| <b>Conclusion</b> | Thus, we have successfully Studied K means algorithm and visualized it. |
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|----------------------------|-----------------------------------------|
| <b>Date of Submission:</b> |                                         |
| <b>Marks out of 10</b>     | <b>Name and Sign of Subject Teacher</b> |
| <b>Remark:</b>             |                                         |